

ALTERNATE EGO: A HYPER-REALISTIC AI-POWERED DIGITAL TWIN PLATFORM

MAJOR PROJECT

*Submitted in the partial fulfillment of the
Requirements for the award of the degree*

of

Bachelor of Technology

in

Computer Science and Engineering

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Ms: Divya Soni



Department of Computer Science & Engineering
Guru Tegh Bahadur Institute of Technology

Guru Gobind Singh Indraprastha University

Dwarka, New Delhi

Year: 2025-2026

DECLARATION

We hereby declare that all the work presented in this Minor Project Report for the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Computer Science & Engineering**, Guru Tegh Bahadur Institute of Technology, affiliated to Guru Gobind Singh Indraprastha University, Delhi is an authentic record of our own work carried out under the guidance of **Ms. Divya Soni**.

Date: 29/04/2026

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CERTIFICATE

This is to certify that the project report entitled “**Alternate Ego: A Hyper-Realistic AI-Powered Digital Twin Platform**”, submitted by **Mr. Prabhdeep Singh, Ms. Simardeep Kaur, Ms. Japeen Kaur and Mr. Aryan Bains** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Computer Science & Engineering**, Guru Tegh Bahadur Institute of Technology, New Delhi is an authentic record of the candidate’s own work carried out by them under our guidance. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Ms. Divya Soni

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Date: 29/04/2026

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Computer Science &
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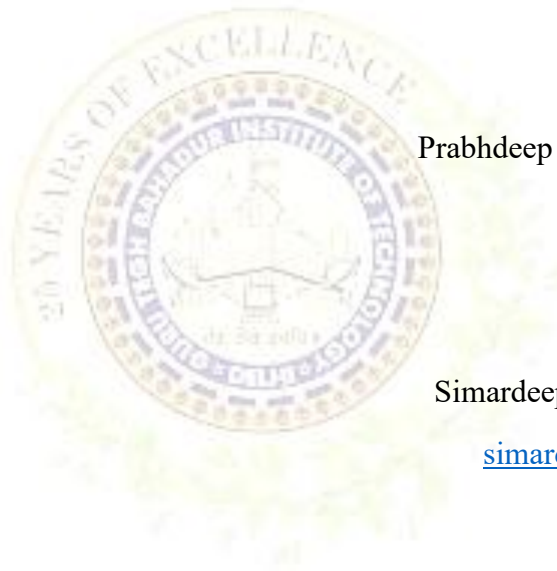


ACKNOWLEDGMENT

We would like to express our gratitude towards our supervisor Ms. Divya Soni for her valuable guidance and frequent suggestions incorporated together with long hours of precious time to help us during this project and for making everything worthwhile and fruitful throughout the project. Without her guidance, we could not have presented this work up to the present standard. She has helped us surmount many hurdles encountered during the course of this project and has been a great source of inspiration.

We would also like to take this opportunity to thank the Hod of Computer Science Department Dr. Ashish Bhardwaj and other faculties who gave us support during this project or in the other aspects of our study at Guru Tegh Bahadur Institute of Technology. Last but not the least, we would like to thank our friends and family for supporting us and giving us the best of guidance.

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ABSTRACT

Alternate Ego is a next-generation AI-powered digital twin platform that enables the creation of hyper-personalized virtual identities capable of replicating an individual's personality, voice, knowledge, and behavioral patterns. The system is built on a privacy-first, fully local architecture, eliminating reliance on cloud APIs and ensuring that all computation, storage, and inference occur on the user's device. This design guarantees zero operational cost and complete data sovereignty while maintaining high-performance, real-time interaction capabilities.

At its core, Alternate Ego employs a Retrieval-Augmented Generation (RAG) framework that transforms heterogeneous user data—including social media content, uploaded datasets, and voice-based interviews—into semantically structured knowledge representations. These inputs are processed through a topic-aware chunking mechanism and embedded into high-dimensional vector spaces using local embedding models. A lightweight, pure Python vector database performs cosine similarity search to retrieve contextually relevant information, which is then fused with dynamic prompt engineering techniques to guide a local large language model in generating accurate, first-person, personality-aligned responses.

The platform integrates a multimodal AI pipeline combining natural language processing, speech processing, and computer vision. Speech-to-text conversion is handled by optimized local inference models, while neural voice cloning enables realistic speech synthesis that mirrors the user's tone and style. Simultaneously, browser-based computer vision models validate and capture emotional expressions to generate dynamic, mood-aware avatars, enhancing the realism of human-AI interaction. The system also incorporates contextual mood detection, adaptive response generation, and interactive conversational memory to simulate authentic human-like engagement.

A defining contribution of Alternate Ego is its end-to-end privacy-preserving data lifecycle. User data is encrypted at rest, processed into non-reversible embeddings, and automatically deleted after ingestion, ensuring that no sensitive raw information is retained or transmitted externally. This approach addresses critical concerns in AI systems related to data security, surveillance, and ethical AI deployment.

From an infrastructure perspective, the platform is engineered using a scalable and modular stack, including FastAPI for backend services, Next.js for an interactive frontend, SQLite for lightweight relational storage, and a local AI runtime for model orchestration. Despite operating entirely offline, the system achieves efficient performance and near real-time response generation through optimized pipelines and resource-aware model loading strategies.

In essence, Alternate Ego represents a paradigm shift in AI system design by combining local-first computing, multimodal intelligence, and personalized knowledge modeling to deliver a secure, scalable, and human-centric digital twin experience.

LOG REPORT

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Details of the Student

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Broader Area: College

Name of Mentor: Ms, Divya Soni

Designation and Contact:

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Student Performance Record

Week	No. of days Scheduled for the Internship	Number of days actually attended	Tasks scheduled for the student	Tasks actually covered by the Student (To be filled by Mentor at organization)	Signature of Mentor
Week 1 12 Jan – 16 Jan	5	5	Research and study of existing digital twin platforms, open-source AI tools, and the original Ego project codebase	1. Studied the original Ego project architecture and identified paid API dependencies. 2. Researched free/local alternatives for each paid service (ElevenLabs → Coqui XTTS, OpenAI → Ollama). 3. Prepared a comparative analysis of 8 paid vs free technology replacements.	

Week 2 19 Jan – 23 Jan	5	5	Environment setup, technology stack finalization, and project architecture design	1. Installed Python 3.10+, Node.js 18+, Ollama runtime, and all development tools. 2. Designed the multi-service architecture: FastAPI backend + Next.js frontend + Ollama AI runtime 3. Created the project directory structure with modular folders: api/, rag/, voice/, avatar/, mcp/, db/ security/.	
Week 3 26 Jan – 30 Jan	5	5	Backend server foundation setup using FastAPI with proper configuration and middleware	1. Initialized FastAPI backend with Uvicorn ASGI server and enabled CORS middleware. 2. Implemented environment configuration using pydantic-settings and .env file management 3. Set up automatic Swagger/OpenAPI documentation at /docs endpoint	
Week 4 2 Feb – 6 Feb	5	5	Database design and implementation using SQLite with Pydantic model.	1. Designed and implemented 5-table SQLite schema: users, twins, conversations, messages, onboarding_sessions. 2. Created full relationship tracking: user → twin → conversation → messages with foreign keys. 3. Built Pydantic request/response models for all API endpoints in db/models.py.	
Week 5 9 Feb – 13 Feb	5	5	RAG (Retrieval-Augmented Generation) engine core development - chunker and embedder modules.	1. Developed chunker.py for topic-based text splitting with source metadata preservation. 2. Implemented embedder.py for generating 768-dimensional vectors via Ollama nomic-embed-text. 3. Pulled and tested nomic-embed-text model (~270MB) and verified embedding quality.	
Week 6 16 Feb – 20 Feb	5	5	RAG engine completion — vector store, prompt builder, and LLM integration	1. Built pure Python JSON vector store with exact cosine similarity math (replaced ChromaDB due to DLL errors). 2. Implemented prompt_builder.py for dynamic system prompts with RAG context injection. 3. Integrated llama3.1:8b via Ollama for personality-aware response generation in llm.py.	

Week 7 23 Feb - 27 Feb	5	5	Social media scraping, data upload pipeline, and transcript processing	1. Implemented public profile scraping using BeautifulSoup + DuckDuckGo search in scrape_processor.py. 2. Built data export upload handler for Instagram/LinkedIn .zip/.json 3. Implemented privacy-first pipeline: Upload → Encrypt → Parse → Feed to RAG → DELETE raw file.	
Week 8 2 Mar - 6 Mar	5	5	Avatar generation system using computer vision and Gemini API integration	1. Implemented webcam capture flow for 4 emotion photos: neutral, happy, sad, angry. 2. Integrated face-api.js with TensorFlow.js models for real-time facial expression validation in browser. 3. Connected Gemini API (gemini-2.0-flash-exp) for generating mood-synced AI cartoon avatars.	
Week 9 9 Mar - 13 Mar	5	5	Voice pipeline development — Speech-to-Text and Text-to-Speech implementation	1. Integrated faster-whisper for local CPU-based speech-to-text transcription in stt.py. 2. Implemented Coqui XTTS v2 voice cloning with lazy model loading to conserve RAM in tts.py. 3. Built Microsoft Edge TTS (edge-tts) as automatic fallback when XTTS fails or runs OOM.	
Week 10 16 Mar - 20 Mar	5	5	Chat system, mood detection, and MCP slash command support	1. Developed RAG-powered chat endpoint: message → retrieval → LLM generation → response + citations. 2. Implemented keyword-based mood detection from reply text (happy, sad, angry, neutral). 3. Built dynamic avatar expression switching based on detected mood in responses. 4. Created /slash command parser and simulated social media posting for LinkedIn and Twitter.	
Week 11 23 Mar - 27 Mar	5	5	Next.js frontend development — landing page, layout, and core components	1. Built the landing page with full-screen cinematic Hero video and glassmorphism design system. 2. Implemented root layout with Playfair Display + Inter fonts, purple/indigo gradients, dark theme. 3. Created reusable components: NameInput, PrivacyBanner, DataUpload, CameraCapture	

Week 12 30 Mar - 3 Apr	5	5	Frontend completion — onboarding flow, chat page, and API integration	1. Built onboarding page with 4-emotion photo capture workflow and webcam integration. 2. Developed split-screen chat page: avatar panel (left) + chat message window (right). 3. Created GeneratingScreen, ScrapingScreen, AvatarView, ChatWindow, and AudioPlayer.	
Week 13 6 Apr - 10 Apr	5	5	Security, encryption, and privacy features implementation	1. Implemented encryption.py with Fernet encryption/decryption for all uploaded files. 2. Built privacy API endpoints for data deletion (/delete-all/{twin_id}) and data summary. 3. Configured auto-generated encryption keys stored locally with secure key management. 4. Developed secure deletion feature: wipes photos, voice samples, uploads, and vector store data completely.	
Week 14 13 Apr - 17 Apr	5	5	End-to-end integration testing, bug fixing, and performance optimization	1. Conducted full end-to-end integration testing: name entry → photo capture → voice interview → chat. 2. Fixed ChromaDB DLL crash by replacing with pure Python JSON vector store with identical cosine math. 3. Optimized lazy model loading for XTTS v2 and faster-whisper to reduce startup memory usage. 4. Resolved CORS middleware issues and API routing conflicts between frontend and backend services.	
Week 15 20 Apr - 24 Apr	5	5	Final documentation, project report, deployment preparation, and presentation	1. Wrote comprehensive README.md with STAR format (Situation-Task-Action-Result) documentation 2. Prepared project demonstration video showcasing the complete digital twin creation and chat flow 3. Completed Software Requirement Specification (SRS), system design, and test plan documents 4. Created final project presentation slides with architecture diagrams and live demo preparation	

(Signature of the student)

Any comments or suggestions for the student performance during the Internship program (to be filled by Mentor at the Organization)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 1		
Day	Date	Work Done (Record point wise)
Monday	12-Jan-2026	Reviewed Alternate Ego project statement and listed main modules: onboarding, RAG, avatar, voice, chat and privacy.
Tuesday	13-Jan-2026	Studied uploaded ZIP structure, README and implementation guide; FastAPI backend and Next.js frontend modules.
Wednesday	14-Jan-2026	Prepared development plan, module-wise task breakdown and notes on digital twin, RAG, voice cloning and avatar generation.
Thursday	15-Jan-2026	Set up local project folders, checked package/requirements files and verified backend/frontend execution requirements.
Friday	16-Jan-2026	Drafted initial architecture notes and finalized weekly scope for onboarding, RAG and digital twin conversation flow.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 2		
Day	Date	Work Done (Record point wise)
Monday	19-Jan-2026	Created FastAPI app structure with routers, middleware and health-check endpoint for the backend service.
Tuesday	20-Jan-2026	Designed database entities for user, twin, onboarding session, conversation and messages.
Wednesday	21-Jan-2026	Implemented onboarding start flow for user/twin/session creation and progress status tracking.
Thursday	22-Jan-2026	Built frontend landing, upload, login and signup layouts with navigation for the user journey.
Friday	23-Jan-2026	Configured environment settings, CORS policy and static storage paths for uploaded media and generated files.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 3		
Day	Date	Work Done (Record point wise)
Monday	26-Jan-2026	Implemented topic-based chunking for profile data, uploaded exports and voice transcripts.
Tuesday	27-Jan-2026	Added embedding-generation flow and vector-store structure for storing twin-specific memory chunks.
Wednesday	28-Jan-2026	Implemented cosine-similarity retrieval and top-k context selection for chat queries.
Thursday	29-Jan-2026	Built scrape/data-export processor to convert public profile and uploaded data into RAG-ready text chunks.
Friday	30-Jan-2026	Added source tracker and prompt builder so generated responses can include relevant source context.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 4		
Day	Date	Work Done (Record point wise)
Monday	02-Feb-2026	Designed onboarding screens for four emotion photographs: neutral, happy, sad and angry.
Tuesday	03-Feb-2026	Added webcam capture logic and frontend validation messages for each expression step.
Wednesday	04-Feb-2026	Implemented backend photo upload/storage endpoint and avatar folder structure per twin.
Thursday	05-Feb-2026	Added avatar generator and emotion-photo selection logic for chat display.
Friday	06-Feb-2026	Tested onboarding progress updates and fixed photo-count/status handling bugs.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology
Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 5		
Day	Date	Work Done (Record point wise)
Monday	09-Feb-2026	Designed voice interview flow with personality-based questions for collecting speech samples.
Tuesday	10-Feb-2026	Implemented browser audio recording and upload flow for voice answers.
Wednesday	11-Feb-2026	Integrated STT transcription service and stored transcript text with question metadata.
Thursday	12-Feb-2026	Added TTS/voice-generation module and voice manager for reference audio handling.
Friday	13-Feb-2026	Processed interview transcripts into personality traits and RAG chunks for digital twin memory.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 6		
Day	Date	Work Done (Record point wise)
Monday	16-Feb-2026	Built split-screen chat layout with avatar panel and message conversation area.
Tuesday	17-Feb-2026	Connected chat API with conversation creation, message saving and retrieval of previous history.
Wednesday	18-Feb-2026	Integrated RAG retrieval and dynamic prompt builder into LLM response generation.
Thursday	19-Feb-2026	Added source-citation formatting and display for fetched memory chunks.
Friday	20-Feb-2026	Implemented mood detection and mapped response mood to avatar expression/photo changes.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 7		
Day	Date	Work Done (Record point wise)
Monday	23-Feb-2026	Implemented slash-command parser for /linkedin and /twitter command formats.
Tuesday	24-Feb-2026	Added MCP/social-poster simulation for post and tweet actions with formatted responses.
Wednesday	25-Feb-2026	Built privacy endpoints for data summary and complete data deletion by twin ID.
Thursday	26-Feb-2026	Integrated encryption and secure-delete utilities for uploaded files and generated media.
Friday	27-Feb-2026	Performed end-to-end testing from onboarding to chat; fixed API routing and validation issues.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 8		
Day	Date	Work Done (Record point wise)
Monday	02-Mar-2026	Polished frontend styling including glassmorphism UI, gradients, fonts and component spacing.
Tuesday	03-Mar-2026	Reviewed codebase, corrected lint/build issues and verified demo screenshots/workflow.
Wednesday	04-Mar-2026	Prepared final documentation for architecture, AI models, API interface and security mechanism.
Thursday	05-Mar-2026	Evaluated system performance, limitations and future enhancement points.
Friday	06-Mar-2026	Prepared final presentation, logbook records and demo checklist for project submission.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 9		
Day	Date	Work Done (Record point wise)
Monday	9-Mar-2026	Built STT using faster-whisper with format conversion and testing.
Tuesday	10-Mar-2026	Developed XTTS voice cloning with lazy loading setup.
Wednesday	11-Mar-2026	Added fallback TTS, voice manager, and audio handling.
Thursday	12-Mar-2026	Designed interview system and integrated voice-to-RAG flow.
Friday	13-Mar-2026	Tested voice pipeline, cloning quality, and memory usage.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 10		
Day	Date	Work Done (Record point wise)
Monday	16-Mar-2026	Built RAG chat API with memory and structured responses.
Tuesday	17-Mar-2026	Added mood detection, avatar switching, and TTS.
Wednesday	18-Mar-2026	Implemented slash commands and social integrations.
Thursday	19-Mar-2026	Integrated memory APIs and validated chat responses.
Friday	20-Mar-2026	Tested chat performance and documented APIs.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 11		
Day	Date	Work Done (Record point wise)
Monday	23-Mar-2026	Initialized Next.js frontend with UI and theme.
Tuesday	24-Mar-2026	Built landing page, inputs, and upload system.
Wednesday	25-Mar-2026	Implemented validation, animations, and consent flow.
Thursday	26-Mar-2026	Developed API client and backend integration.
Friday	27-Mar-2026	Completed layout, animations, and navigation testing.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

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Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 12		
Day	Date	Work Done (Record point wise)
Monday	30-Mar-2026	Built camera capture with emotion detection.
Tuesday	31-Mar-2026	Developed voice interview with recording flow.
Wednesday	1-Apr-2026	Integrated onboarding with multi-step UI.
Thursday	2-Apr-2026	Created chat UI with avatars and messages.
Friday	3-Apr-2026	Integrated TTS and completed frontend testing.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology
Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 13		
Day	Date	Work Done (Record point wise)
Monday	6-Apr-2026	Implemented file encryption and key handling.
Tuesday	7-Apr-2026	Built privacy APIs for data control.
Wednesday	8-Apr-2026	Added secure deletion and cleanup flow.
Thursday	9-Apr-2026	Tested encryption and data removal.
Friday	10-Apr-2026	Conducted security audit and documentation.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 14		
Day	Date	Work Done (Record point wise)
Monday	13-Apr-2026	Performed integration testing and found bugs.
Tuesday	14-Apr-2026	Fixed CORS, upload, and chat issues.
Wednesday	15-Apr-2026	Optimized model loading and added streaming.
Thursday	16-Apr-2026	Improved performance and error handling.
Friday	17-Apr-2026	Completed regression and performance testing.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

Guru Tegh Bahadur Institute of Technology

Department of Computer Science and Engineering

Major Project / Internship Log-Book

Week: 15		
Day	Date	Work Done (Record point wise)
Monday	20-Apr-2026	Documented README and project metrics.
Tuesday	21-Apr-2026	Completed SRS, design, and API docs.
Wednesday	22-Apr-2026	Prepared demo video, slides, and screenshots.
Thursday	23-Apr-2026	Finalized report and formatting.
Friday	24-Apr-2026	Delivered presentation and submitted project.
Remarks by Project Guide/Mentor:		

(Name & Signature of Mentor at the Organization along with Seal)

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